

Andrés Riveros Valdevenito

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Employment & Education

Postdoctoral Research Associate , Department of Mathematical Sciences, Carnegie Mellon University.	Sep 2026 - Present
PhD in Statistics , Columbia University . • Co-advised by Professor Marcel Nutz & Professor Philip Protter.	Sep 2020 - May 2026
Master of Philosophy in Statistics , Columbia University. • As part of the PhD program.	Sep 2020 - May 2024
Master of Arts in Statistics , Columbia University. • As part of the PhD program.	Sep 2020 - Feb 2023
Master of Engineering Sciences, minor in Applied Mathematics , Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile. • With Maximum Distinction. • Advised by Professor Jaime San Martín.	Mar 2019 - Jun 2020
Mathematical Engineer Professional Degree , Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile. • With Maximum Distinction.	Mar 2014 - Jun 2020
Bachelor of Engineering Sciences in Mathematics , Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile. • With Distinction.	Mar 2014 - Dec 2018

Work Experience

Dept. of Applied Mathematics, Universidad de Granada , Visiting Researcher. • Invited by Professor Alejandro Gárriz. • Project related to optimal transport theory.	Granada, Spain Jun 2025
Center of Mathematical Modelling, Universidad de Chile , Research Intern. • Under guidance of Professor Joaquín Fontbona. • Research review of state-of-the-art Hamiltonian Monte Carlo methods.	Santiago, Chile Jan 2019
Superintendency of Banks and Financial Institutions (SBIF, now Financial Market Commission, CMF) , Analytics Intern. • Under Patricio Jaramillo, director of the area of modelling and methodology.	Santiago, Chile Jan 2018
Assarlab, Faculty of Medicine, Universidad de Chile , Research Intern. • Under professor Rodrigo Assar. • Created algorithms based on previous research to analyze the learning process of fish.	Santiago, Chile Jan 2017

Honours & Awards

Columbia University: • Dean's Fellow, Graduate School of Arts & Sciences.	Sep 2020
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- Excellence Scholarship for National Science Masters Degree Students, Chile.

Publications

Linear Convergence of Gradient Descent for Quadratically Regularized Optimal Transport.	2026+
A. González-Sanz, M. Nutz, A. Riveros Valdevenito , Preprint, arXiv:2509.08547  .	
Polyak–Łojasiewicz Inequality for Quadratically Regularized Optimal Transport	2026
A. González-Sanz, M. Nutz, A. Riveros Valdevenito , SIAM Journal on Optimization (To appear), arXiv:2605.27175  .	
Monotonicity in Quadratically Regularized Linear Programs.	2025
A. González-Sanz, M. Nutz, A. Riveros Valdevenito , SIAM Journal on Optimization, Vol 35, N°2, pp 1419-1437, arXiv:2408.07871  .	
Markov Jump Times and their Cox Construction.	2024
P. Protter, A. Riveros Valdevenito , Annals of Operations Research, arXiv:2310.07003  .	
On the Guyon-Lekeufack Volatility Model.	2024
M. Nutz, A. Riveros Valdevenito , Finance & Stochastics, Vol 28, N°4, pp 1203-1223, arXiv:2307.01319  .	
Propiedad de Martingala, Volatilidad Estocástica & Burbujas Financieras	2020
A. Riveros Valdevenito , Master's Thesis, advised by professor Jaime San Martín, available at repositorio.uchile  .	

Invited & Contributing Talks

- *Minghui Conference*, Columbia University, April 2026, April 2025, April 2024, April 2023.
- *IV Workshop on Stochastic Analysis and Optimization, applied to economics, finance, energy and insurance*, Puerto Natales, Chile, December 2025.
- *9th Eastern Conference in Mathematical Finance* (Poster), CMU, October 2025.
- *Optimal Transport Reading Group*, CMU, October 2025.
- *Probability Seminar*, FCFM, Universidad de Chile, August 2025.
- *Young Researchers Seminar*, Universidad de Granada, June 2025.
- *Statistics and Optimal Transport Workshop* (Poster), Columbia University, March 2025.
- *Statistics Student Seminar*, Columbia University, January 2025.
- *23rd Northeastern Probability Seminar*, Columbia University, November 2024.
- *22nd Northeastern Probability Seminar*, NYU, November 2023.
- *Oxford Statistics Student Seminar*, June 2020.

Service

- Co-organizer of the CMU Probability/Financial Math seminar, Fall 2026.
- **Referee for:** Finance & Stochastics, Mathematics of Operations Research, Operations Research, SIAM Journal of Mathematical Analysis.

Technologies & Skills

Basic Programming: Python, R, Java, Unix, SQL, Visual Basic, Matlab, SciLab, Latex, MS Office.

Mathematical Programs: STATA, Stan, AMPL, BOCOP, FreeFem.

Languages: Fluent in English and Spanish. Basic French.

Teaching

- **Lecturer at Carnegie Mellon University :** Probability (21325-A) Sep 2026 - Present
- **Teaching Assistant at Columbia University :** Stochastic Methods in Finance STAT GR4265 (Fall 2025), Probability Qualifying Exam Review Session Teacher (Summer 2023, 2024, 2025), PhD Probability II STAT GR6302 (Spring 2024, Spring 2025, Spring 2026), PhD Probability I STAT GR6301 (Fall 2023, Fall 2024), Core Competency Exam Review Session Teacher (Spring 2024), Probability STAT S4203/W5203 (Summer 2023), Introduction to Statistics with Calculus STAT S1201/W1201 (Summer 2022, Spring 2023), Linear Regression Models STAT S4205/S5205 (Fall 2022), Introduction to Statistics STAT W1101 (Spring 2022, Fall & Spring 2021), Introduction to Statistical Reasoning STAT W1001 (Summer 2021, Fall 2020). Sep 2020 - May 2026
- **Teaching Assistant at Universidad de Chile, FCFM:** Supervisor of Entire Core Courses of Mathematics (2019,2020,2021), Stochastic Calculus MA5402 (Spring 2018, 2019), Markov Processes MA4401 (Fall 2018), Introduction to Calculus MA1001 (Fall 2017), Linear Algebra MA1102 (Spring 2016): Grader & Test Supervisor, Introduction to Calculus MA1001 (Fall 2016): Grader & Test Supervisor. Mar 2014 - Mar 2022